FEA Membranes — Verification & Validation Record

Native Windows application: element library, constraints and model operations

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Scope and method

This document records the automated verification performed on the *FEA Membranes* native application: the Quad4 and Quad8 plane-stress membrane elements, axial bar and XY-decoupled spring elements, RBE2 rigid constraints, boundary-condition handling, stress recovery, and the model operations (meshing, merging, deletion) that must preserve a valid, solvable model.

Every case below is implemented as an automated test in `tests/FeaCore.Tests/SolverTests.cs` and executes on every build:

```
dotnet test tests/FeaCore.Tests/FeaCore.Tests.csproj
```

A failing case fails the build, so this record stays current by construction.

Conventions. Plane stress throughout; the application's world coordinates are X–Y with Y drawn downward on screen (sketches here use conventional Y-up orientation for readability). Test values use a consistent lb–in–psi set. "Exact" means the discretised problem has a closed-form solution that the element must reproduce to the stated tolerance — these are *verification* (patch-test) cases, not approximations.

Contents

1 Element verification

1.1 E1 — Quad4 uniaxial tension (single-element patch case)

Test: `Quad_UniaxialTension_ExactForBilinear`
Basis: single 10×10 element, $E = 10^7$, $\nu = 0$, $t = 0.1$, total $P = 1000$ lb on the free edge. Closed form: $\sigma_x = P/(t\,w) = 1000$ psi, $u = \sigma L/E = 1.0 \times 10^{-3}$ in.
Acceptance: u, σ_x, σ_{VM} exact to 10^{-6} (relative); $\Sigma R_x = -1000$ to 10^{-6}.

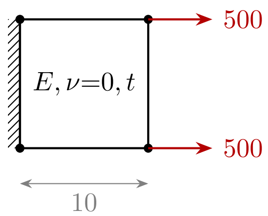

Figure 1: E1: single Quad4, clamped edge, uniform end load. $\sigma_x = 1000$ psi exact.

1.2 E2 — Quad4 plate far-field stress (mesh-level sanity)

Test: `Plate_20x20_FarFieldStress`
Basis: 100×100 plate, 20×20 Quad4 mesh, $\nu = 0.3$, one edge clamped, uniform load on the far edge. By Saint-Venant, mid-plate $\sigma_x = 2100/(100 \cdot 0.1) = 210$ psi.
Acceptance: mid-plate element average within 2%.

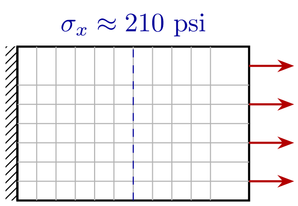

Figure 2: E2: clamped plate under end tension; far-field stress checked at mid-plate.

1.3 E3 — Quad4 constant-stress patch state through nodal averaging

Test: `NodalStresses_UniformField_EqualElementStress`
Basis: 2×2 mesh with *consistent* edge loads (corner nodes carry half the mid-node value: 250/500/250 for 1000 total). A constant stress field must be reproduced exactly at every corner-extrapolated, node-averaged evaluation point.

Acceptance: $\sigma_x = 1000$, $\sigma_y = 0$, $\sigma_{VM} = 1000$ at all nine averaged nodes (4 decimals); contributing-element counts verified (interior node 4, edge 2, corner 1).

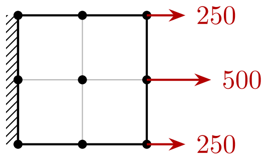

Figure 3: E3: consistent loads for uniform traction; averaging must not disturb a constant field.

1.4 E4 — Quad8 uniaxial patch test

Test: `Quad8_PatchTest_UniaxialTensionExact`
Basis: single 8-node serendipity element. Consistent nodal loads for uniform traction on a quadratic edge are $\frac{1}{6} : \frac{4}{6} : \frac{1}{6}$ of the edge total. Same closed form as E1.
Acceptance: $u = 10^{-3}$ at all free-edge nodes (9 decimals); σ_x exact at the centre and at *all eight* averaged node positions (4 decimals); reaction sum exact.

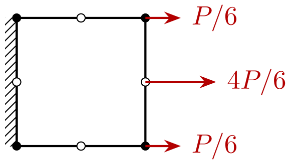

Figure 4: E4: Quad8 patch test (open circles are midside nodes) with consistent quadratic-edge loads.

1.5 E5 — Quad8 pure in-plane bending (exact linear stress)

Test: `Quad8_PureBending_ExactLinearStress`
Basis: a *single* Quad8 (20×10, mid-plane at $y = 0$) under pure bending applied as a consistent linear end traction $\sigma_x(y) = 200\,y$. The quadratic displacement field carries linear strain exactly — the defining capability a Quad4 lacks (shear locking).
Acceptance: $\sigma_x = 200\,y$ reproduced at every node (3 decimals).

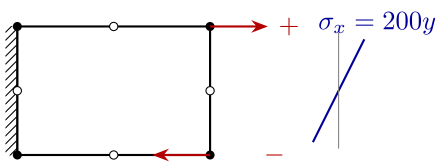

Figure 5: E5: single Quad8 in pure bending; the recovered stress profile is exactly linear.

1.6 E6 — Quad8 quarter-point (Barsoum crack-tip) configuration

Test: `Quad8_QuarterPoint_CrackTipConfigurationSolves`
Basis: the two midside nodes adjacent to a "crack-tip" corner are moved to the quarter points. With the fully isoparametric formulation, the geometry mapping then embeds the $1/\sqrt{r}$ strain singularity used for stress-intensity-factor extraction. Gauss points are interior, so assembly remains well-defined; the singularity sits exactly at the tip node.
Acceptance: assembles ($\det J > 0$ at all Gauss points), solves, all displacements and averaged stresses finite, ΣR_y balances the applied load to 5 decimals.

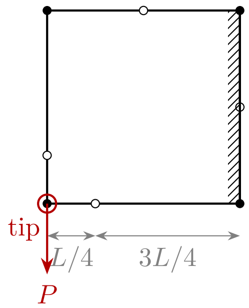

Figure 6: E6: midside nodes shifted to the quarter points adjacent to the tip corner.

1.7 E7 — Bar axial stiffness ($\delta = PL/EA$)

Test: `Bar_AxialElongation_PLOverEA`
Basis: $E = 10^7$, $A = 0.1$, $L = 10$, $P = 1000 \Rightarrow \delta = 0.01$. A second, orthogonal bar provides transverse restraint and must carry ≈ 0 force.
Acceptance: δ exact to 10^{-9}; $P = +1000$ (tension-positive), $\sigma = 10{,}000$ psi; transverse bar force $\leq 10^{-6}$.

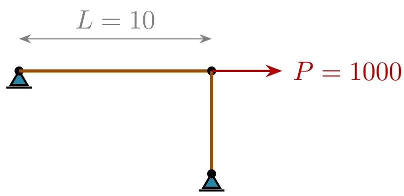

Figure 7: E7: axial bar with transverse support bar; elongation and force sign verified.

1.8 E8 — Spring element (XY-decoupled fastener idealisation)

Test: `Spring_DecoupledXY_ForceAndDisplacement`
Basis: the spring applies independent stiffness k in X and Y (it is deliberately *not* axial — this is the sheet load-transfer fastener idealisation). $F = k\delta$: $k = 10^5$, $F = 100 \Rightarrow \delta = 10^{-3}$.
Acceptance: δ and the recovered spring force exact to 10^{-9}.

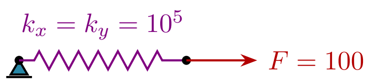

Figure 8: E8: decoupled spring; force–displacement relation exact per direction.

2 Boundary conditions and constraints

2.1 C1 — Enforced displacement is exact (direct elimination)

Test: `EnforcedDisplacement_IsExact`
Basis: prescribed $d_x = 2 \times 10^{-3}$ on the free edge of the E1 plate $\Rightarrow \sigma_x = E\varepsilon = 2000$ psi. Prescribed values are applied by direct elimination (partitioned solve), so there is *no* penalty-number approximation.
Acceptance: prescribed displacement reproduced to 12 decimals; σ_x to 10^{-3}.

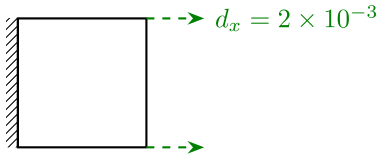

Figure 9: C1: enforced edge displacement (dashed green); stress matches $E\varepsilon$ exactly.

2.2 C2 — RBE2 ties selected DOFs exactly

Test: `Rbe2_TiesSelectedDofsExactly`
Basis: RBE2 is enforced as an *exact multipoint constraint*: tied DOFs share one solve unknown (no penalty). Here the loaded edge of a plate is tied in X only, with the load applied to a single node of the group.
Acceptance: all tied d_x identical to 12 decimals; d_y demonstrably independent (Poisson contraction differs); the reaction sum carries the full load through the tie.

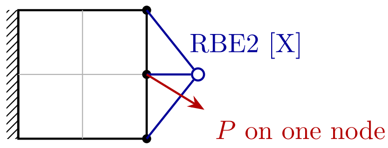

Figure 10: C2: edge tied in X by an RBE2 spider; load on one node moves the whole group.

2.3 C3 — Constraint applied through the independent node

Test: `Rbe2_ConstrainsViaIndependentNode`
Basis: a BC anywhere in a tie group holds the group, and loads on dependent nodes act on the group. A bar (PL/EA) carries the tied pair; the closed form must be recovered.
Acceptance: group displacements identical (12 decimals); $\delta = PL/EA$ to 9 decimals; reaction sum exact.

2.4 C4 — Conflicting prescribed values inside a tie group

Test: `Rbe2_ConflictingPrescribedValues_Throw`
Basis: two different enforced displacements within one tie group are physically contradictory.
Acceptance: a clear solver error — never a silently averaged or garbage answer.

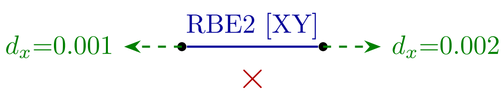

Figure 11: C4: contradictory enforced values in one rigid group are rejected.

3 Solution diagnostics (error-trapping verification)

3.1 D1 — Orphan node diagnostic

Test: `OrphanNode_ThrowsWithNodeId`
Basis: a node with no attached stiffness makes K singular; a pre-solve check names the offending node(s) instead of failing opaquely.
Acceptance: the error message contains the orphan node id.

3.2 D2 — Under-constrained model detection (residual guard)

Test: `UnderConstrained_ThrowsInsteadOfSilentGarbage`
Basis: with rigid-body modes present, the sparse LU factorisation does *not* necessarily throw — it can return zeros. Every solution is therefore checked against $\|Ku - f\| \leq 10^{-6}\,\|f\|$.
Acceptance: a loaded but unconstrained plate raises a clear "add constraints" error.

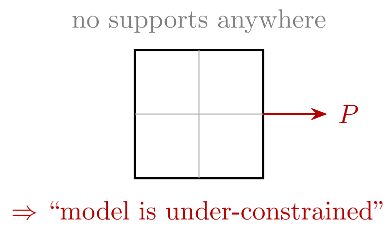

Figure 12: D2: rigid-body modes are caught by the post-solve equilibrium residual check.

4 Stress recovery (nodal averaging)

4.1 S1 — Constant field invariant under averaging

Test: `NodalStresses_UniformField_EqualElementStress` (shared with E3). A constant stress state must survive corner extrapolation and node averaging unchanged at every node.

4.2 S2 — Step change averages to the mean at shared nodes

Test: `NodalStresses_StepChange_AveragesAtSharedNodes`
Basis: two elements in series with different thickness override: $\sigma_x = P/(t\,w)$ gives 500 psi in the thick element and 1000 psi in the thin one. At the shared interface nodes the average of the two corner values, 750 psi, must be reported.
Acceptance: per-node values 500 / 750 / 1000 to 3 decimals.

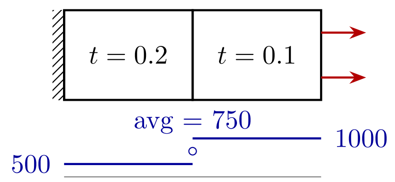

Figure 13: S2: thickness step; the averaged nodal value at the interface is the exact mean.

5 Model-operation verification

These cases verify that meshing and editing operations always leave a valid, solvable model: element references intact, ids unique, no dangling entities. Cases with significant physical content are sketched.

ID	Test	What it proves
M1	`Mesher_FlatQuad_NodeAndElementCounts`	Structured-mesh counts; corner FE nodes coincide w geometry; re-mesh *fully replaces* the old mesh (uni ids, valid references); result solves
M2	`Mesher_ArcEdge_MatchesWebtoolSample`	Circular-arc edge geometry matches the webtool mes to 4 decimals (cross-implementation consistency)
M3	`Quad8_Mesher_CountsAndMidsides`	Serendipity grid count $(2M+1)(2N+1) - MN$; m sides exactly at edge midpoints; Quad4↔Quad8 re-m switches cleanly
M4	`Mesher_MoveGeometryPoint_RemeshesAffectedSurfaces`	Geometry edits re-mesh affected surfaces with stored visions; mesh follows geometry; unmeshed surfaces touched
M5	`Mesher_DeleteElements_RemovesOrphanNodes`	Element deletion removes newly unreferenced nodes
M6	`Mesher_DeleteNodes_CascadesToAttachedEntities`	Node deletion cascades to e ments/springs/bars/RBE2s; remaining referen intact
M7	`Mesher_DeleteSurface_RemovesMeshAndOrphanGeometry`	Surface deletion removes its mesh, attached entities, shared geometry points
M8	`Mesher_ClearMesh_KeepsNodesSharedWithOtherSurfaces`	After stitching, clearing one surface keeps seam no the other still references
M11	`Mesher_MergeCoincidentNodes_NegativeCoordinates`	Coincidence search correct for negative / orig straddling coordinates
M12	`Mesher_AddSurface_SharedPoints`	Snapped corner picks share geometry points; movin shared point re-meshes both surfaces; duplicate corn rejected
M13	`Mesher_SpringPointGrid_CountsAndSpacing`	Grid counts, pitch, collapsed single-row/column haviour
M16	`Rbe2_CleanupOnMeshAndMergeOperations`	RBE2s pruned on re-mesh; references remapped thro merges

Table 1: Model-operation integrity cases (tabulated).

5.1 M9 — Coincident-node merge stitches two meshes into one plate

Test: `Mesher_MergeCoincidentNodes_StitchesTwoSurfacesIntoContinuousPlate`
Basis: two 5×10 surfaces meshed separately share the $x = 5$ line as duplicated seam nodes. After merging, the pair must behave as one continuous 10×10 plate: uniaxial tension gives the exact bilinear answer.
Acceptance: 3 seam nodes merged; every element keeps valid node references; $\sigma_x = 1000$ psi in *every* element (4 decimals); reactions balance.

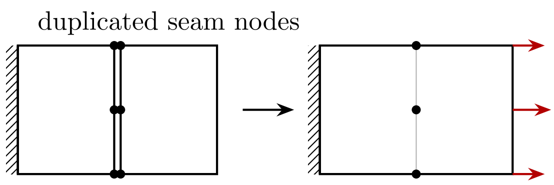

Figure 14: M9: merge within tolerance fuses the seam; the stitched pair recovers the exact one-plate answer.

5.2 M10 — Merge never destroys spring-joined fastener pairs

Test: `Mesher_MergeCoincidentNodes_PreservesSpringPairsAndBcs`
Basis: coincident nodes joined by a spring are the intentional fastener idealisation; merging them would silently delete the modelled load transfer. Plain coincident pairs do merge; BCs carry to the surviving node; zero-length bars created by a merge are removed; a selection-scoped merge leaves out-of-scope nodes untouched.

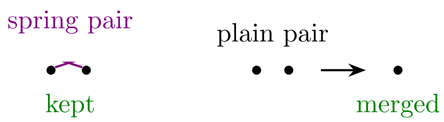

Figure 15: M10: spring-connected coincident pairs are exempt from merging.

5.3 M14/M15 — Springs at spring points: exactly-two rule and visibility scope

Tests: `Mesher_SpringsAtSpringPoints_ExactlyTwoVisibleNodesRule`, `Mesher_SpringsAtSpringPoints_Onl`
Basis: at each spring point, a spring is created only when *exactly two visible* FE nodes lie within the coincidence range (one spring per point; duplicates skipped; > 2 in range is skipped, not guessed). With three coincident layers nothing is created until one layer is hidden — then the remaining pair connects, and no spring touches the hidden layer.

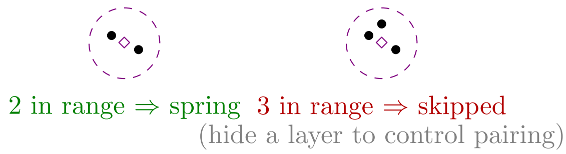

Figure 16: M14/M15: the exactly-two-visible-nodes rule for automatic fastener creation.

6 System level and file integrity

ID	Test	What it proves
F1	`SampleModel_LoadsAndSolves`	A real webtool-exported model (curved edge, BCs, bars) loads, solves, satisfies equilibrium (ΣR_x to 4 decimals), all displacements finite and bounded
F2	`WebtoolJson_RoundTrip`	Webtool JSON compatibility: camelCase fields, edge radii, BCs, springs, bars survive load → save → load

Table 2: System-level cases.

7 Independent cross-checks performed during development

Not part of the automated suite, recorded for completeness:

- The webtool's solver (the same element formulation, independently implemented in JavaScript) passes the equivalents of E1, E7, E8, C1, D1 and E2 in its own headless Node.js harness — two independent implementations agreeing on the same closed-form answers.
- The native and webtool meshers produce numerically identical meshes for the same curved-edge model (M2 pins this permanently).

8 Known gaps — recommended additions for a formal validation suite

Verification above is largely on rectangular, undistorted elements with closed-form targets. A formal validation suite should add:

1. **Distorted-element patch tests** (Irons): constant-stress patch on irregular Q4/Q8 assemblies. The isoparametric formulation should pass; it is currently untested in distorted form.

2. **Convergence vs handbook solutions**: plate with central circular hole ($K_t \to 3.0$ under refinement); Q8 vs Q4 convergence rates on the same problem. (Model `stress-conc-hole.json` in this folder is the candidate geometry.)
3. **Quarter-point SIF benchmarks** (once extraction lands): centre- and edge-cracked plates vs handbook K_I (Tada; Rooke & Cartwright), with mesh-refinement sensitivity.
4. $\nu \neq 0$ **single-element checks**: biaxial case with Poisson coupling (current single-element exactness cases use $\nu = 0$).
5. **Shape-sensitivity sweep**: accuracy vs aspect ratio and skew, to set modelling guidance.
6. **Large-model conditioning**: solve time and residual-check headroom at 10^5+ DOF.
7. **Reactions under non-zero enforced displacement** vs closed form.
8. **Fastened-doubler load transfer** end-to-end vs an independent solution (e.g. Nastran or closed-form one-dimensional joint theory). (Models `fastened-doublers*.json` in this folder are the candidates.)